

ONE-WAVE FRAMEWORK

Book 1 — Micro

Chapter 5: The Neutron — Two-Shell Pressure Balance

Version: 3.0 Date: July 1, 2026 Class: B — Applied Layer Spine: Gray / 2D / 3D / Mathematics / Predictions / Yellow Audit / Future Work / Closing Thoughts

Dependencies: C-08 Spin-half, D-05 Harmonic Shell, E-03 Surface, E-04 Coupling, B-01 Balance, B-06b Four Views, CCD-02, C-317 Boundary-Tension Weave, C-318 Four-Interaction Mass-Effect Response Status: YELLOW (two-shell model parked pending CCD-02)

Gray — Standard Model Reference

The neutron is a composite particle: two down quarks and one up quark, bound by gluons. Mass: 939.565 MeV/c². Charge: 0. Spin: 1/2. The neutron is slightly heavier than the proton (by ~1.3 MeV/c²). Outside the nucleus, the neutron is unstable — it decays into a proton, electron, and antineutrino with a half-life of approximately 611 seconds (~10 minutes). Inside the nucleus, neutron stability is maintained by the nuclear force. The neutron has a negative charge radius squared ($r_n^2 = -0.1161 \text{ fm}^2$) — meaning its charge distribution has a negative outer shell and positive inner shell. This is directly measurable via neutron-electron scattering.

Standard Model strength: neutron structure well described by QCD. Standard Model limitation: neutron charge radius structure (negative outer, positive inner) requires a specific quark distribution that is not simply derived. The neutron decay mechanism (beta decay) requires the weak force — a fourth separate force.

2D One-Wave Interpretation

In 2D, the neutron is a two-shell braided loop.

Like the proton, it is a three-vortex braid. Unlike the proton, the inner and outer pressure shells have opposite signs. The net charge is zero — but not because there is no charge structure. It is because the positive inner shell and negative outer shell cancel.

In 2D cross-section: Inner ring: positive pressure (expression outward from core). Outer ring: negative pressure (compression inward from boundary). The two rings are in balance — $B = 0$ at equilibrium (B-01).

The two-shell structure is the neutron's defining feature. The proton has one dominant shell (positive throughout). The neutron has two shells that cancel.

3D One-Wave Interpretation

In 3D, the neutron is a two-shell Persistent Mode.

Inner shell: $R_+ = 0.7331 \text{ fm}$ — positive pressure region (expressive, outward) Outer shell: $R_- = 0.8409 \text{ fm}$ — negative pressure region (compressive, inward) Shell width: $\sigma = 0.210 \text{ fm}$

The two shells are held in balance by surface tension (E-03) and coupling (E-04). The net charge is zero because the positive inner shell and negative outer shell produce equal and opposite pressure cushions (B-06b).

The inner shell's outward pressure = Expression (B-03) The outer shell's inward pressure = Compression (B-04) The boundary between them = the Mirror Gate at nuclear scale (C-01)

This is the four views (B-06b) operating at nuclear scale: Inward = outer shell compressing inward
 Outward = inner shell expressing outward Across = the boundary surface between them Over = the
 roll-off at the outer boundary producing the net-zero charge field

Neutron decay: Outside the nucleus, the balance between inner and outer shells is unstable. The
 outer shell loses coupling with neighbors (no neighboring nucleons to couple to). Coupling loss:
 β_i drops. The outer shell begins to collapse. As the outer shell collapses, the inner expression
 pressure is released. The released inner pressure = the electron (charge, spin-half from roll-off).
 The coupling loss event itself = the neutrino (low-coupling mode, deferred to Ch 9). The remaining
 inner-shell mode = the proton.

This is neutron beta decay without the weak force. It is a boundary collapse event producing three
 modes from one unstable two-shell structure.

CCD-02 Block: The full derivation of the neutron shell model requires $E_{opp}(R)$ — the opposing
 pressure function between the two shells. This is blocked pending the proton boundary derivation
 in Chapter 2 / Book 2.

Mathematics

Two-shell model (parked pending CCD-02): Inner shell: $R_+ = 0.7331$ fm Outer shell: $R_- = 0.8409$ fm
 Shell width: $\sigma = 0.210$ fm

Calibration reanalysis (D-407): - R_+ and R_- are profile-center radii; σ is a profile width. - σ
 is not a second estimate of adjacent shell spacing. - If, and only if, the two centers are adjacent
 harmonic shells, the nearest integer assignment is $n_+ = 7$, $n_- = 8$ with $\lambda = 0.659395$ fm. - *The*
fitted radii differ from that 7/8 model by about 0.2% each. - *The fit provenance, uncertainties, and*
physical adjacency derivation are absent, so λ is a conditional sensitivity-test value, not a
 measured nuclear wavelength.

These values pass charge form factor sanity checks internally: The measured neutron charge radius
 squared $r_n^2 = -0.1161$ fm² is consistent with a negative outer shell at R_- and positive inner shell
 at R_+ .

Charge neutrality condition (B-01): $B = 0 \Rightarrow (k_{EE_inner} + k_{II_inner}) - (k_{RR_outer} + k_{LL_outer}) = 0$

Inner shell pressure (expression): $P_{inner} \sim K_p * |\nabla \psi_{inner}|^2$ (E-02)

Outer shell pressure (compression): $P_{outer} \sim K_p * |\nabla \psi_{outer}|^2$ (E-02)

Balance: $P_{inner} = P_{outer} \Rightarrow \text{net charge} = 0$

Shell surface tension (E-03): $E_{inner} = \sigma * 4\pi R_+^2$ $E_{outer} = \sigma * 4\pi R_-^2$

Coupling between shells (E-04): $E_c = (1/2)a\psi_{inner}^2 + (1/2)b\psi_{outer}^2 + c\psi_{inner}\psi_{outer}$

Decay condition (sketch — Yellow): When $\beta_i(\text{outer})$ drops below coupling threshold: $T_{outer} < T_{break}$ (B-09) $\Rightarrow$ outer shell break condition $\Rightarrow$ inner expression pressure released as electron
 $\Rightarrow$ coupling-loss event as neutrino $\Rightarrow$ remaining inner shell = proton

Predictions

1. Neutron charge radius squared r_n^2 is negative because the outer shell is negative. Prediction:
 $r_n^2 = \int \rho(r) * r^2 dr < 0$ when outer shell dominates at large r . This matches the
 measured value -0.1161 fm². One-Wave gives a geometric explanation for what QCD explains
 through quark distribution.

2. The neutron-proton Mass-Effect difference must be produced by how the outer shell changes the complete four-interaction recurrence. The added shell changes electrical-shell stress, Mirror-Gate pressure, Boundary-Tension confinement, internal geometry, and their cross-couplings. Prediction:

$$\Delta m_{np,\text{eff}} = \frac{1}{3} \text{Tr}(\mathcal{M}_n - \mathcal{M}_p),$$

using one fixed work metric for both states. Shell energy alone is not divided by c^2 and treated as the mechanism.

3. Neutron decay is a boundary collapse event, not a weak force interaction. Prediction: neutron decay rate outside the nucleus correlates with the coupling strength β_i of the outer shell to neighboring modes. In vacuum: no neighbors $\Rightarrow$ coupling drops $\Rightarrow$ decay occurs. Inside nucleus: neighbors maintain coupling $\Rightarrow$ neutron is stable.
4. No weak force needed for neutron decay. The decay products (proton, electron, neutrino) emerge from the two-shell collapse geometry, not from a W-boson exchange. Prediction: the energy and momentum distribution of beta decay products matches the two-shell collapse model without W-boson.
5. The neutron is the same three-vortex braid as the proton with an additional outer compression shell. Prediction: neutron and proton have identical inner structure. The difference is entirely the outer shell.

Yellow Audit

- Full two-shell model derivation blocked by CCD-02 ($E_{\text{opp}}(R)$)
- $\kappa_{\pm}$ (equilibrium stiffness) not yet derived
- Decay condition mathematics is sketch level only
- Neutrino as coupling-loss event deferred to Chapter 8
- Neutron and proton four-interaction response tensors have not yet been derived from one fixed coefficient set
- W-boson replacement claim is structural — formal derivation deferred

Future Work

Resolve CCD-02: derive $E_{\text{opp}}(R)$ from proton boundary work (Book 2). Complete two-shell model with $\kappa_{\pm}$. Derive neutron decay rate from outer shell coupling loss. Compare predicted beta decay energy distribution to experimental data. Derive the neutron-proton Mass-Effect difference from the two stable four-interaction profiles without per-object refitting.

Closing Thoughts

The neutron's negative charge radius is one of the most interesting puzzles in nuclear physics. It directly tells you the charge is not uniformly zero — it has a positive inside and negative outside. The Standard Model explains this through quark distribution. One-Wave explains it through geometry: two shells with opposite pressure signs.

The neutron decay mystery — why does it need the weak force? — also dissolves. The outer shell loses its coupling when it has no neighbors. The shell collapses. The pieces come out. No W-boson. No fourth force. Just a boundary collapse.

END OF BOOK 1 CHAPTER 5 One wave. Mirror builds. Mark Wright. Kitty Hawk V0.